

Sujith S

Support Engineer | Application Engineering · Process Automation · AWS · Unix/Linux · Debugging & Root Cause Analysis

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Summary

Application and support engineer currently pursuing Int.MSc Computational Statistics and Data Analytics at VIT University (2022-2027), with hands-on experience building, deploying, and maintaining scalable backend services and REST APIs on AWS. Skilled at troubleshooting and debugging technical systems, writing process automation scripts in Python and Unix/Linux environments, monitoring production services, and resolving operational incidents. Experienced translating business requirements into concrete technical deliverables, collaborating with cross-functional teams, and driving operational efficiency through SLA definition, metrics monitoring, and root cause analysis. Published researcher with a track record of owning end-to-end technical solutions from design through production maintenance.

Education

Int.MSc Computational Statistics and Data Analytics

VIT University | 2022-2027

Coursework: Machine Learning, Deep Learning, Statistical Modeling, NLP, Data Structures & Algorithms, Distributed Systems, Database Systems

CS50 Programs

Harvard CS50 (Online)

Completed [CS50x](#), [CS50P](#), [CS50AI](#), [CS50SQL](#), and [CS50R](#) covering algorithms, AI, Python, and full-stack systems.

Skills

Application & Support Engineering: Troubleshooting, Debugging, Root Cause Analysis, Incident Management, Alarm Monitoring, On-Call Support, SLA Definition, Operational Metrics, Service Health Monitoring, Bug Fixes, Process Automation

Languages & Scripting: Python, SQL, Bash/Shell Scripting, Java, R, JavaScript, TypeScript

Cloud & Infrastructure: AWS (Lambda, S3, SageMaker, Redshift, EKS), Docker, CI/CD Pipelines, Linux/Unix, Prometheus

Backend & APIs: FastAPI, REST APIs, PostgreSQL, Redis, Apache Kafka, Apache Airflow, MLflow, Delta Lake

Frameworks & Tools: PyTorch, TensorFlow, LangChain, LangGraph, CrewAI, Scikit-learn, XGBoost, Apache Spark, dbt, FAISS

Work Experience

Founder

[Shoot the Breeze](#) |

2023 - Present

- Owned end-to-end development and operational maintenance of ML-powered backend services on AWS; implemented CI/CD pipelines, Docker-based deployments, and Prometheus monitoring to ensure service health and rapid incident response.
- Wrote Python and shell automation scripts to streamline repetitive operational tasks, reduce manual toil, and improve system reliability; performed root cause analysis on production failures and drove permanent fixes to prevent recurrence.
- Translated business and functional requirements into scalable technical deliverables; defined SLAs, tracked operational metrics, and maintained documentation for all services and processes.

App/Web Developer

[CropSky](#) | VIT-TBI Research & Innovation Hub |

2024 - Present

- Built and maintained scalable REST APIs and ML-powered backend services on cloud infrastructure; collaborated with engineering teams to troubleshoot application issues, resolve incidents, and onboard business teams onto platform features.
- Developed automated data pipelines querying large-scale structured databases; monitored service health, investigated operational anomalies, and coordinated cross-functional fixes to maintain platform stability for production sellers.

Projects

Demand Forecasting System

[GitHub](#)

Python, FastAPI, PostgreSQL, Redis, MLflow, Docker, AWS, Prophet, XGBoost

- Designed and deployed production-grade REST inference APIs on AWS with staging-to-production CI/CD pipelines; implemented MLflow experiment tracking, automated model retraining with drift detection, and real-time monitoring dashboards to maintain service health and SLA adherence.

Multi-Agent Business Automation System

[GitHub](#)

Python, CrewAI, LangGraph, LangChain, FastAPI, PostgreSQL, Redis, Docker, AWS, Prometheus

- Architected a scalable agentic automation platform with Prometheus-based alarm monitoring and structured escalation workflows; debugged multi-step agent failures, performed root cause analysis on task routing errors, and maintained operational health of enterprise-scale automated workflows.

Recommendation Engine

[GitHub](#)

Python, PyTorch, FAISS, FastAPI, PostgreSQL, Redis, Docker

- Engineered and maintained a hybrid recommendation service with optimized low-latency inference APIs; implemented online model update pipelines, monitored service performance metrics, and resolved latency and accuracy issues through systematic debugging and root cause analysis.

Research & Publications

Stock Market Analysis and Prediction Using Machine Learning

2025

[El Ruha International Conference on Scientific Research](#), pp. 730-740. Designed end-to-end ML pipelines applying LSTM, GRU, Transformer, ARIMA, XGBoost, LightGBM, and RL agents (DQN, PPO, A2C); built feature engineering pipelines, evaluated 21 models, and distilled findings into a peer-reviewed publication. [\[GitHub\]](#)

Certifications & Achievements

- [Google Generative AI Leader](#)
- [Google Cloud Computing Foundations](#)
- [FreeCodeCamp Legacy Full Stack](#)
- [OCI 2025 Certified Developer Professional](#)
- AWS [Solutions Architect](#) and [Developer](#)
- [Software Engineer Certification - HackerRank](#)